E-COMMERCE CUSTOMER BEHAVIOR & SALES ANALYSIS

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Analysis Period: January 2023 – March 2024

Dataset: E-Commerce Customer Behavior Dataset

Data Analytics Portfolio Project

1.EXECUTIVE SUMMARY

This project analyzes an e-commerce customer behavior and sales dataset containing 5,000 transactions recorded between January 2023 and March 2024. The analysis was conducted using Microsoft Excel to evaluate sales performance, customer purchasing patterns, product category performance, payment preferences, and customer satisfaction.

The dataset was examined and validated through data quality checks, followed by exploratory analysis using PivotTables, calculated metrics, and interactive visualizations. The analysis identified key performance indicators, including total revenue of 4,915,544.57, 5,000 orders, 11,100 units sold, and an average order value of 983.11.

The findings highlight Electronics as the leading revenue-generating category, Istanbul as the highest-revenue city, and credit cards as the most frequently used payment method. Returning customers accounted for 59.8% of all orders, while 71.56% of customers gave a rating of four or five stars.

The dashboard and supporting analysis provide a clear overview of business performance and customer behavior, helping identify patterns and areas that may deserve further investigation.

2. PROJECT OVERVIEW

The objective of this project is to analyze e-commerce sales data and customer behavior to identify meaningful business patterns and performance indicators.

The analysis focuses on the following objectives:

- Evaluate overall sales performance through revenue, order volume, quantity sold, and average order value.
- Identify the product categories and cities generating the highest revenue.
- Examine customer purchasing behavior across payment methods and device types.
- Analyze customer satisfaction using customer ratings.
- Compare new and returning customers based on their contribution to sales and engagement indicators.
- Assess delivery performance and identify variations across cities.

The project uses a structured Excel workflow, combining data validation, exploratory analysis, PivotTables, calculated metrics, and dashboard visualizations to communicate the results clearly.

3. DATASET DESCRIPTION

The dataset used in this project is an e-commerce customer behavior dataset containing 5,000 transaction records and 18 columns. It covers the period from January 2023 to March 2024.

Each record represents an order and includes information about customers, products, transactions, and shopping behavior.

The dataset contains the following groups of variables:

- **Order and Customer Information:** Order_ID, Customer_ID, Age, Gender, and City.
- **Product and Sales Information:** Product_Category, Unit_Price, Quantity, Discount_Amount, and Total_Amount.
- **Transaction Details:** Date and Payment_Method.
- **Customer Behavior:** Device_Type, Session_Duration_Minutes, Pages_Viewed, and Is_Returning_Customer.
- **Delivery and Satisfaction:** Delivery_Time_Days and Customer_Rating.

The dataset includes multiple product categories, payment methods, device types, and cities, allowing the analysis to explore different aspects of sales performance and customer behavior.

The analysis covers January 2023 to March 2024. Therefore, the 2024 results represent only the first three months of the year and should not be interpreted as a full-year comparison.

4. TOOLS AND METHODOLOGY

Tools Used

- **Microsoft Excel:** Used for data preparation, validation, exploratory analysis, PivotTables, calculated metrics, charts, and dashboard development.
- **Excel Tables:** Used to structure the working dataset and support dynamic formulas and analysis.
- **PivotTables and PivotCharts:** Used to summarize data and compare sales and customer behavior across different dimensions.

Methodology

The project followed a structured analytical workflow:

1. **Data Inspection:** Reviewed the dataset structure, column names, record count, and data types.

2. **Data Preparation:** Created a separate working copy while preserving the original raw dataset.
3. **Data Quality Assessment:** Checked missing values, duplicate Order_IDs, invalid values, logical consistency, and returning-customer values. Data types were also reviewed manually.
4. **Exploratory Data Analysis:** Examined sales, product categories, cities, payment methods, devices, customer ratings, and delivery performance.
5. **KPI Calculation:** Calculated key indicators, including total revenue, total orders, quantity sold, total discounts, and average order value.
6. **Data Visualization:** Developed charts and an interactive Excel dashboard to present the main findings.
7. **Reporting:** Consolidated the results into a structured analysis report.

This workflow was designed to maintain data consistency, support reliable analysis, and present the findings in a clear and accessible format.

5. DATA QUALITY ASSESSMENT

Data quality checks were performed to assess the reliability and consistency of the working dataset before conducting the analysis. Six checks were documented, covering missing values, duplicate order identifiers, data types, invalid values, logical consistency, and returning-customer values.

All six checks passed based on the recorded results. The data type review was conducted manually, while the remaining checks used Excel formulas.

Quality Check	Description	Result
Missing Values	Number of empty cells	0 – Pass
Duplicate Order IDs	Duplicate order identifiers	0 – Pass
Data Types	Expected data types reviewed manually	Pass – Manual
Invalid Values	Negative or impossible values	0 – Pass
Logical Consistency	Total amount calculation consistency	0 – Pass
Returning Customer Values	Valid TRUE/FALSE values	Pass – 0 invalid

Data Quality Notes

The duplicate check focused specifically on Order_ID values and did not assess duplicate records across all columns. The data type check was performed manually. The validation results indicate that no missing values, duplicate order identifiers, or invalid values were detected in the checks performed.

6. KEY PERFORMANCE INDICATORS (KPIs)

The following key performance indicators provide an overview of the overall e-commerce business performance during the analysis period.

Key Performance Indicator	Value
Total Revenue	4,915,544.57
Total Orders	5,000
Quantity Sold	11,100
Total Discounts	124,264.02
Average Order Value	983.11

The dataset contains 5,000 orders with a total revenue of 4,915,544.57 and 11,100 units sold. The average order value is 983.11, while total discounts amount to 124,264.02.

These indicators establish a baseline for evaluating sales performance and provide context for the detailed findings presented in the following sections.

7. KEY BUSINESS FINDINGS

7.1 Sales Performance by Product Category

The analysis of sales by product category reveals significant differences in revenue contribution. Electronics generated the highest revenue, reaching 2,328,806.81, representing approximately 47.37% of total revenue. Home & Garden ranked second with 908,348.86, followed by Sports with 754,563.56.

In terms of quantity sold, Sports recorded the highest volume at 1,519 units, while Books recorded the lowest at 1,291 units. These results show that the categories generating the highest revenue are not necessarily those with the highest sales volume.

Product Category	Revenue	Quantity Sold
Beauty	156,584.74	1,424
Books	72,744.52	1,291
Electronics	2,328,806.81	1,379
Fashion	375,214.93	1,411
Food	96,138.67	1,355
Home & Garden	908,348.86	1,353
Sports	754,563.56	1,519
Toys	223,142.48	1,368
Grand Total	**4,915,544.57**	**11,100**

7.2 Sales Performance by City

The city-level analysis shows that Istanbul generated the highest revenue, with 1,334,122.56 across 1,284 orders, representing approximately 27.14% of total revenue. Ankara ranked second with 657,535.82, followed by Izmir with 567,534.67.

Eskisehir recorded the lowest revenue at 178,978.43 across 210 orders. These results highlight the differences in revenue contribution across the cities included in the dataset.

City	Revenue	Orders
Adana	427,059.63	378
Ankara	657,535.82	735
Antalya	315,549.89	374
Bursa	459,076.31	496
Eskisehir	178,978.43	210
Gaziantep	318,408.41	349
Istanbul	1,334,122.56	1,284
Izmir	567,534.67	600
Kayseri	312,303.91	257
Konya	344,974.94	317
Grand Total	**4,915,544.57**	**5,000**

7.3 Sales Performance by Payment Method

The payment method analysis shows that Credit Card was the most frequently used payment method, accounting for 2,012 orders and generating revenue of 1,916,670.38. Debit Card ranked second with 1,265 orders and revenue of 1,250,502.73.

Digital Wallet generated 963,056.98 across 965 orders, while Bank Transfer accounted for 510 orders and revenue of 550,872.17. Cash on Delivery was the least frequently used method, with 248 orders and revenue of 234,442.31.

Overall, Credit Card and Debit Card together represented approximately 65.54% of all orders.

Payment Method	Revenue	Orders
Bank Transfer	550,872.17	510
Cash on Delivery	234,442.31	248
Credit Card	1,916,670.38	2,012
Debit Card	1,250,502.73	1,265
Digital Wallet	963,056.98	965
Grand Total	**4,915,544.57**	**5,000**

7.4 Sales Performance by Device Type

The device type analysis indicates that Mobile devices generated the highest revenue, reaching 2,810,468.64 across 2,795 orders. This represents approximately 57.17% of total revenue and 55.9% of all orders.

Desktop devices accounted for 1,711 orders and generated revenue of 1,650,103.78. Tablet devices recorded the lowest contribution, with 494 orders and revenue of 454,972.15.

These findings show that mobile devices represented the largest share of both order volume and revenue in the dataset.

Device Type	Revenue	Orders
Desktop	1,650,103.78	1,711
Mobile	2,810,468.64	2,795
Tablet	454,972.15	494
Grand Total	**4,915,544.57**	**5,000**

7.5 Customer Satisfaction Analysis

The customer rating analysis indicates a generally positive satisfaction pattern. Out of 5,000 orders, 1,833 received a five-star rating and 1,745 received a four-star rating. Combined, these ratings account for 71.56% of all recorded ratings.

A total of 759 orders received three stars, while 429 received two stars and 234 received one star. The results show that four- and five-star ratings represent the majority of customer feedback in the dataset.

Customer Rating	Number of Orders	Percentage
1 Star	234	4.68%
2 Stars	429	8.58%
3 Stars	759	15.18%
4 Stars	1,745	34.90%
5 Stars	1,833	36.66%
Grand Total	**5,000**	**100.00%**

7.6 New vs. Returning Customers Analysis

The customer segmentation analysis shows that returning customers accounted for 2,990 orders, representing 59.8% of all orders, while new customers accounted for 2,010 orders (40.2%).

Returning customers generated revenue of 2,948,020.84, compared with 1,967,523.73 from new customers. Average customer ratings were very similar, at 3.90 for returning customers and 3.91 for new customers.

These results indicate that returning customers contributed a larger share of both order volume and revenue, while customer ratings remained nearly identical between the two groups.

Customer Type	Number of Orders	Revenue	Average Rating
New Customers	2,010	1,967,523.73	3.91
Returning Customers	2,990	2,948,020.84	3.90
Grand Total	**5,000**	**4,915,544.57**	**3.90**

7.7 Delivery Performance Analysis

The delivery performance analysis shows an average delivery time of 6.50 days across all orders. A total of 3,401 orders (68.02%) were delivered within seven days, while 595 orders (11.90%) required more than ten days.

At city level, Konya recorded the shortest average delivery time at 5.95 days, whereas Istanbul recorded the longest at 6.67 days. The difference between these two cities was 0.72 days.

These findings provide an overview of delivery speed and highlight variations in average delivery time across the cities included in the dataset.

City	Average Delivery Time (Days)
Adana	6.61
Ankara	6.55
Antalya	6.42
Bursa	6.40
Eskisehir	6.63
Gaziantep	6.28
Istanbul	6.67
Izmir	6.59
Kayseri	6.26
Konya	5.95
Overall Average	**6.50**

7.8 Monthly Revenue Analysis

The monthly revenue analysis shows fluctuations in sales performance throughout the analysis period. January 2024 recorded the highest monthly revenue at 396,874.92, followed by August 2023 with 391,505.89. April 2023 recorded the lowest monthly revenue at 257,332.58.

Revenue performance varied from month to month, with several months recording relatively strong results. However, the 2024 data covers only January to March, so comparisons between 2023 and 2024 should be interpreted carefully rather than treated as full-year comparisons.

Month	Revenue
January 2023	286,694.32
February 2023	338,784.92
March 2023	321,716.01
April 2023	257,332.58
May 2023	351,902.03
June 2023	324,356.96
July 2023	346,618.03
August 2023	391,505.89
September 2023	262,602.86
October 2023	352,286.61
November 2023	282,606.10
December 2023	384,717.52
January 2024	396,874.92
February 2024	342,913.06
March 2024	274,632.76

7.9 Customer Demographics Analysis

The customer demographics analysis examines sales performance across different age groups and gender categories.

Among the age groups, customers aged 35–44 generated the highest revenue, reaching 1,483,006.07 across 1,521 orders. The 25–34 age group followed with revenue of 1,436,981.67 across 1,476 orders. Together, these two groups accounted for 2,997 orders, representing 59.94% of all orders.

Regarding gender, female customers generated 2,474,009.26 in revenue, while male customers generated 2,381,028.70. The Other category accounted for 60,506.61. Overall, revenue contributions from female and male customers were relatively close.

Table 1. Sales by Age Group

Age Group	Orders	Revenue
Under 25	998	1,005,207.55
25–34	1,476	1,436,981.67
35–44	1,521	1,483,006.07
45–54	769	747,805.60
55+	236	242,543.68
Grand Total	**5,000**	**4,915,544.57**

Table 2. Sales by Gender

Gender	Orders	Revenue
Female	2,492	2,474,009.26
Male	2,435	2,381,028.70
Other	73	60,506.61
Grand Total	**5,000**	**4,915,544.57**

8. RECOMMENDATIONS

Based on the findings of this analysis, several recommendations can be considered to support sales performance, improve customer experience, and strengthen business operations.

The following recommendations are derived from the patterns observed in the dataset and should be considered as areas for further investigation and potential improvement.

1. Strengthen Electronics Category Performance

Electronics generated the highest revenue among all product categories. The business could consider maintaining product availability, monitoring demand, and evaluating promotional strategies to support continued performance in this category.

2. Focus on High-Revenue Cities

Istanbul recorded the highest revenue contribution among the cities analyzed. The business could examine the factors behind this performance and assess whether similar sales strategies could be applied in other cities with lower revenue contributions.

3. Optimize Mobile Shopping Experience

Mobile devices accounted for the largest share of both revenue and orders. The business should continue monitoring the mobile shopping experience and consider improving website usability, navigation, and checkout processes to support customer convenience and reduce potential purchasing barriers.

4. Maintain Customer Satisfaction

The analysis shows that 71.56% of orders received ratings of four or five stars. The business should continue monitoring customer feedback, identify recurring issues in lower-rated orders, and use this information to improve products and services.

5. Encourage Customer Retention

Returning customers accounted for 59.8% of all orders and generated a larger share of total revenue. The business could consider customer loyalty initiatives, personalized offers, and targeted communication to encourage repeat purchases and strengthen long-term customer relationships.

6. Monitor Delivery Performance

The average delivery time was 6.50 days, with some differences across cities. The business should monitor delivery times regularly, investigate the reasons behind longer delivery periods, and explore opportunities to improve delivery efficiency and customer experience.

9. CONCLUSION

This project provided an overview of e-commerce sales performance and customer behavior through the analysis of 5,000 transaction records covering January 2023 to March 2024.

Using Microsoft Excel, the dataset was reviewed, validated, and analyzed through exploratory analysis, PivotTables, calculated metrics, and interactive visualizations. The analysis identified key performance indicators and highlighted important patterns across product categories, cities, payment methods, device types, customer ratings, and customer segments.

The findings showed that Electronics generated the highest revenue, Istanbul recorded the highest city-level revenue, and mobile devices accounted for the largest share of sales. Returning customers also contributed a greater proportion of orders and revenue, while customer ratings were generally positive.

Overall, this project demonstrates the use of Excel for transforming raw transaction data into structured analysis and meaningful business insights. It also highlights how data analytics can support performance monitoring, customer understanding, and informed business decisions.

10. DASHBOARD OVERVIEW

The interactive Excel dashboard summarizes the key findings of the e-commerce analysis. It presents the main performance indicators and visualizes sales trends, product category performance, city-level revenue, payment methods, device usage, customer ratings, and customer segmentation.

The dashboard provides a consolidated view of business performance and allows users to explore the results through interactive visualizations.

